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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	3	Academic Year :	2026-27

Task - 3

Aim

To integrate external RESTful APIs into a React application using asynchronous HTTP requests (fetch / axios) within the useEffect lifecycle hook, implementing robust data rendering, client-side search filtering, loading state skeletons, and defensive error handling boundaries.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. Asynchronous fetch from GitHub REST API (https:

```
//api.github.com/users/Dharmikvaghani/repos).
2. Defensive state management handling loading skeletons, HTTP error statuses, and empty search arrays.
3. Real-time text filter allowing dynamic querying across repository names and topics.
// src/pages/Projects.jsx
function Projects() {
  const [repos, setRepos] = useState([]);
  const [loading, setLoading] = useState(true);
  const [search, setSearch] = useState('');
  useEffect(() => {
    fetch('https://api.github.com/users/Dharmikvaghani/repos')
      .then(res => res.json())
      .then(data => { setRepos(data); setLoading(false); })
      .catch(err => { console.error(err); setLoading(false); });
  }, []);
  const filtered = repos.filter(r => r.name.toLowerCase().includes(search.toLowerCase()));
  return (
    <div className="projects-container">
      <input value={search} onChange={e => setSearch(e.target.value)} placeholder="Filter..." />
      {loading ? <Skeleton /> : filtered.map(r => <RepoCard key={r.id} repo={r} />)}
    </div>
  );
}
Output
```

Output

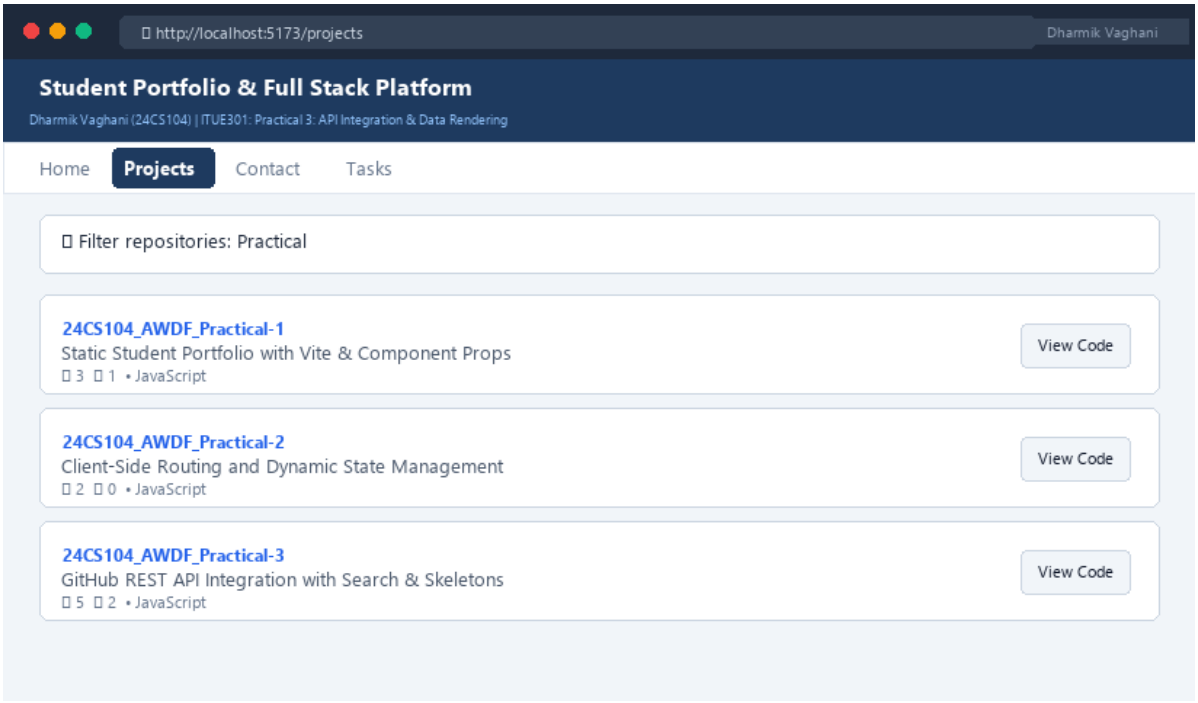


Figure 3: Execution Output & Functional Verification for Practical 3